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The most surprising discovery for me was learning how images get represented as numerical data, not just as "pictures." Before this lab, I pictured images as plain visuals, but then I found out every image is built from thousands— or even millions of pixels, and each pixel holds numerical values that describe color plus brightness. When I saw the image shown like arrays of numbers, it helped my brain grasp how computers interpret visual information, in a more concrete way. The mathematical operations we put into place really changed what the image looked like in the end. For example simple actions, like tweaking pixel values, could shift brightness, adjust contrast, or alter color intensity. And then there were other operations, such as applying filters and transformations that made different visual effects by rearranging or processing groups of pixels based on specific mathematical rules. It was kind of interesting to notice how straightforward calculations can still cause noticeable changes across the whole image, not only in one spot. The most difficult technique for me to wrap my head around was image filtering and how kernels work. At first it was hard to visualize how a tiny matrix could sharpen something, blur it, or detect edges inside an image. But after experimenting with different filters and just watching what happened, I started to get it, the math operations either emphasize or suppress certain visual features depending on the kernel and the way it's applied.

The image processing bits we did in lab clarified some of the hidden or underlying steps that AI image systems might do before generating or editing anything. Even though the models can crank out impressive results on their own, this lab also made it clear that there are basic image manipulations first, and those are what lets a lot of the newer applications work. I can already picture a bunch of real-world uses for these approaches. Such as image enhancement showing up in digital photography, medical imaging, security systems, autonomous vehicles, and even social media. Edge detection paired with filtering helps highlight key objects inside a scene, while adjusting color and brightness can make the image more usable, better seen, or less confusing. For a future project, I think it would be interesting to blend classic image processing with AI methods. The traditional steps could clean up and prepare the images ahead of time, before they get analyzed by machine learning models. That combo might push accuracy upward, while also lowering the workload the AI system must carry.

The part of image processing that grabbed me most is how computers can spot and understand visuals. I keep thinking about how those mathematical operations can reshape images and how that links to more advanced computer vision systems that can detect faces, objects, and even repeat shapes or motifs. It feels like a key foundation for today's AI applications. This lab changed the way I look at digital photography and image editing, because it made it clear that every small edit is really based on messing with numerical pixel values. Things that look simple in a photo editor are, in practice, the result of lots of calculations happening "behind the curtains". I honestly feel I have more respect for the tech behind image creation and enhancement, because it's not magic, it's mostly structured number work. One thing I still wonder is how modern AI image generation models build older, more traditional image processing ideas. I want to dig into how neural networks analyze images, and then somehow create realistic new content using what they learned from existing visual information. Exploring the bridge between classical techniques and modern AI seems like an exciting next step for me, even if I'm still wrapping my head around how the pieces connect.